PeerLearn: System Test Report (System Validation)

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Result: 20 of 20 system test cases passed, 0 failed. Every case in Test Plan v1.1, section 9, was run against the complete, running system, the way a user reaches it: through nginx over HTTPS to the API and the PostgreSQL database. No defects were found.

1. Test environment

Item	Value
Date of run	2 Nov 2026, 08:17 to 08:40 IST (TC-01 to TC-19); TC-20 on 31 Oct 2026, 10:09 IST in GitHub Actions
Code version	`main` branch, commit `32855fc` (github.com/amoghshetty05/peerlearn)
System under test	nginx with `infra/nginx.conf` (HTTPS on TLS 1.2 and 1.3, HTTP redirect, `/api` proxy) serving the built React frontend; Node.js 22 API with `NODE_ENV=production` and `ENFORCE_HTTPS=true`; PostgreSQL 16 with the restricted application role
Test data	`seed-admin.js` and `seed-demo.js` (Admin, 2 Learners, 4 Tutors), accounts created by the tests, and `seed-perf.js` (1,000 users including 200 Tutors) for TC-13 and TC-18
Tools	Python requests (API calls over HTTPS), psql (database checks), openssl s_client (TLS), Chromium through Playwright (TC-17), `backend/scripts/load-test.js` (TC-18), GitHub Actions with a kind Kubernetes cluster (TC-20)
Note	The local nginx was version 1.24, which does not accept the `http2 on;` line, so that one line was removed for this run. The Docker image uses nginx 1.27 with the file unchanged.

2. Summary

ID	Scenario	Requirements	Result
TC-01	Valid registration	FR-01	**Pass**
TC-02	Non-university email	FR-01	**Pass**
TC-03	Login success and failure	FR-02, SR-04	**Pass**
TC-04	Role restriction	FR-02, SR-05	**Pass**
TC-05	Subjects and availability	FR-03, FR-04	**Pass**
TC-06	Ranked matches	FR-05, FR-06, FR-07, FR-15	**Pass**

ID	Scenario	Requirements	Result
TC-07	Booking and contact privacy	FR-08, FR-11, FR-14, SR-03	**Pass**
TC-08	Request expiry	FR-08, FR-11	**Pass**
TC-09	Rating rules	FR-09, FR-13	**Pass**
TC-10	Group capacity	FR-10	**Pass**
TC-11	Report and suspend	FR-12	**Pass**
TC-12	Password storage	NFR-01, SR-02	**Pass**
TC-13	Matching speed	NFR-03	**Pass**
TC-14	Rate limit and lockout	NFR-04, SR-08	**Pass**
TC-15	Injection and validation	SR-06	**Pass**
TC-16	Audit log	NFR-06, SR-07	**Pass**
TC-17	Dashboard and responsive layout	FR-13, NFR-05	**Pass**
TC-18	API response time	NFR-02	**Pass**
TC-19	HTTPS and token expiry	SR-01, SR-04	**Pass**
TC-20	Deployed stack on Kubernetes	SR-01, FR-02	**Pass**

3. Detailed results

Each case lists the Test Plan's expected result, then every step that was run and what was actually observed.

TC-01: Valid registration (FR-01): Pass

Expected: 201 Created; the new user can log in; the response has no password

#	Step	Observed	OK
1	POST /api/auth/register with ana.15e63@pes.edu and a 10-character password	HTTP 201; response fields: ['createdAt', 'email', 'id', 'name', 'phone', 'role', 'status']	Yes
2	Log in with the new account	HTTP 200; token returned: True	Yes

TC-02: Non-university email (FR-01): Pass

Expected: 400 VALIDATION_ERROR; no account is created

#	Step	Observed	OK
1	POST /api/auth/register with ana.15e63@gmail.com	HTTP 400 VALIDATION_ERROR: Use your @pes.edu email address.	Yes
2	Count users with that email in the database	0 rows	Yes

TC-03: Login success and failure (FR-02, SR-04): Pass

Expected: Correct: 200 and a token. Wrong: 401 and no token

#	Step	Observed	OK
1	Log in as vikram@pes.edu with the correct password	HTTP 200; token returned: True; role learner	Yes
2	Log in as vikram@pes.edu with a wrong password	HTTP 401 UNAUTHENTICATED; token returned: False	Yes
3	Log out, then reuse the same token	logout HTTP 204; reuse HTTP 401 UNAUTHENTICATED	Yes

TC-04: Role restriction (FR-02, SR-05): Pass

Expected: 403 FORBIDDEN; the report is unchanged

#	Step	Observed	OK
1	Asha (Learner) files a report about a Tutor	HTTP 201; report id 1, status open	Yes
2	PATCH /api/admin/reports/1 with asha's token (Learner)	HTTP 403 FORBIDDEN	Yes
3	PATCH /api/admin/reports/1 with meera's token (Tutor)	HTTP 403 FORBIDDEN	Yes
4	Read the report's status in the database	open	Yes
5	Learner calls the Tutor-only endpoint GET /api/tutors/me	HTTP 403 FORBIDDEN	Yes

TC-05: Subjects and availability (FR-03, FR-04): Pass

Expected: First slot 201; overlapping slot 409 SLOT_OVERLAP; 11th subject rejected with an error

#	Step	Observed	OK
1	Add Monday 16:00 to 17:00	HTTP 201	Yes
2	Add Monday 16:30 to 17:30 (overlaps)	HTTP 409 SLOT_OVERLAP	Yes
3	Boundary: a 29-minute slot, then a 30-minute slot	29 min: HTTP 400 VALIDATION_ERROR; 30 min: HTTP 201	Yes
4	Add 10 subjects	HTTP codes: [201]	Yes
5	Add an 11th subject	HTTP 400 VALIDATION_ERROR: You can teach at most 10 subjects.	Yes

TC-06: Ranked matches (FR-05, FR-06, FR-07, FR-15): Pass

Expected: Exactly 5 Tutors, in descending score, none without Maths, no score above 100, rating bonus applied only to the rated Tutor

#	Step	Observed	OK
1	Test data: Maths tutors plus 1 Physics-only tutor; Nisha (Beginner) given 3 ratings of 5	tutors teaching maths: 8; Nisha's ratings: 3	Yes
2	Asha creates a Maths help request for Mon 09 Nov 16:00 IST	HTTP 201; request id 5	Yes
3	GET /api/requests/{id}/matches	Meera Iyer 100 (bonus 0); Mtutor2 15e63 100 (bonus 0); Nisha Kulkarni 97 (bonus 10); Mtutor0 15e63 93 (bonus 0); Mtutor1 15e63 87 (bonus 0)	Yes
4	Exactly 5 tutors, highest score first	5 tutors; scores [100, 100, 97, 93, 87]	Yes
5	Every tutor teaches Maths; the Physics-only tutor is absent	all teach maths: True; physics tutor listed: False	Yes
6	No score above 100	max score 100	Yes
7	Rating bonus only for the tutor with 3 ratings	bonuses: Meera Iyer: 0, Mtutor2 15e63: 0, Nisha Kulkarni: 10, Mtutor0 15e63: 0, Mtutor1 15e63: 0	Yes

TC-07: Booking and contact privacy (FR-08, FR-11, FR-14, SR-03): Pass

Expected: Before acceptance email and phone are absent; after acceptance they are present; both users receive a notification

#	Step	Observed	OK
1	Asha requests Meera's slot 2026-11-02T16:00:00+05:30	HTTP 201; status pending; expires 2026-11-04T08:17:26+05:30	Yes
2	Asha views Meera's profile before acceptance	HTTP 200; email shown: False; phone shown: False	Yes
3	Meera accepts	HTTP 200; status accepted	Yes
4	Asha views Meera's profile after acceptance	HTTP 200; email meera@pes.edu; phone +91 98450 33333	Yes
5	Vikram (no accepted session with Meera) views her profile	email shown: False; phone shown: False	Yes
6	Read both users' notifications	Meera: "Asha Rao asked for a maths session on Mon, 2 Nov, 04:00 pm. Reply within 48 hours."; Asha: "Meera Iyer accepted your session on Mon, 2 Nov, 04:00 pm. Contact details are now visible."	Yes

TC-08: Request expiry (FR-08, FR-11): Pass

Expected: Status becomes expired and the Learner is notified

#	Step	Observed	OK
1	Vikram requests Rohan's Tuesday 16:00 slot; Rohan does not answer	HTTP 201; status pending; expires 2026-11-04T08:17:27+05:30	Yes
2	Move the request 49 hours into the past in the test database (the 48-hour rule uses real time)	created_at set to 49 hours ago	Yes
3	Wait for the background expiry job (runs every minute)	status after 55 s: expired	Yes
4	Read Vikram's notifications	Your session request expired because the tutor did not answer within 48 hours.	Yes

Note: The 48-hour wait was simulated by moving the request's timestamps back in the test database; the expiry job itself ran unchanged.

TC-09: Rating rules (FR-09, FR-13): Pass

Expected: Pending: rejected. Score 5: 201. Second rating: 409. Score 6: 400

#	Step	Observed	OK
1	Learner books a session starting 08:21 IST and rates it while it is still pending	HTTP 409 CONFLICT: You can rate a session only after it is completed.	Yes
2	Tutor accepts the session	HTTP 200, status accepted	Yes
3	Wait until the start time (08:21 IST) passes	session status: completed	Yes
4	Rate the completed session 5 with a comment	HTTP 201	Yes
5	Rate the same session again	HTTP 409 CONFLICT	Yes
6	Rate with a score of 6	HTTP 400 VALIDATION_ERROR	Yes
7	Tutor's dashboard shows the new rating (FR-13)	HTTP 200; rating: average 5.0 from 1 rating	Yes

TC-10: Group capacity (FR-10): Pass

Expected: First 3 join; the 4th gets 409 GROUP_FULL; after a member leaves the 4th can join

#	Step	Observed	OK
1	Asha creates a Maths group with 3 places	HTTP 201; group id 1	Yes
2	Four learners try to join in turn	results: ['200', '200', '200', '409	Yes

#	Step	Observed	OK
		GROUP_FULL’]	
3	Member 1 leaves, then the 4th learner tries again	leave HTTP 200; join HTTP 200	Yes

TC-11: Report and suspend (FR-12): Pass

Expected: Report is stored; the suspended Tutor cannot log in; the Admin action is logged

#	Step	Observed	OK
1	Vikram reports a Tutor	HTTP 201; report 2 status open	Yes
2	Admin lists open reports	HTTP 200; report 2 listed: True	Yes
3	Admin suspends the Tutor	HTTP 200; report status suspended	Yes
4	Suspended Tutor tries to log in	HTTP 403 ACCOUNT_SUSPENDED: This account has been suspended by a moderator.	Yes
5	Look for the Admin action in the audit log	ADMIN_SUSPEND report=2 user=20	Yes

TC-12: Password storage (NFR-01, SR-02): Pass

Expected: password_hash starts with and cost is 10 or more; no response contains a hash

#	Step	Observed	OK
1	Read the hash prefix of every stored password	20 users; all bcrypt : True; cost factors used: [12]	Yes
2	Call 10 read endpoints and search each response for hashes, password fields or tokens	responses with leaks: none	Yes

TC-13: Matching speed (NFR-03): Pass

Expected: Every response arrives in under 2 seconds

#	Step	Observed	OK
1	Database loaded with the performance data set	1022 users, 211 tutors, 83 teaching maths	Yes
2	Run 20 match requests (GET /api/requests/{id}/matches) through nginx over HTTPS	all 20 returned 5 ranked tutors; fastest 5 ms, average 6 ms, slowest 9 ms	Yes

TC-14: Rate limit and lockout (NFR-04, SR-08): Pass

Expected: The account is locked (423) for 15 minutes; the 101st request gets 429

#	Step	Observed	OK
1	Enter a wrong password 5 times	HTTP codes: [401, 401, 401, 401, 401]	Yes
2	Then log in with the correct password	HTTP 423 ACCOUNT_LOCKED: Too many failed logins. Try again in 15 minutes.	Yes
3	Read how long the lock lasts	about 15 minutes remaining	Yes
4	Send 101 requests in one minute with one user's token	requests 1-100: [200]; request 101: 429	Yes
5	Another user makes a request at the same time	HTTP 200	Yes

TC-15: Injection and validation (SR-06): Pass

Expected: 400 VALIDATION_ERROR or no match; no data is leaked; tables are unchanged

#	Step	Observed	OK
1	Log in with ' OR '1'='1 as email and password	HTTP 401 UNAUTHENTICATED; token returned: False	Yes
2	Register with a SQL-injection email	HTTP 400 VALIDATION_ERROR	Yes
3	Create a help request whose topic is an SQL command	HTTP 201; stored as plain text: "'); DELETE FROM sessions; –"	Yes
4	Create a help request with a 10,000-character topic	HTTP 400 VALIDATION_ERROR	Yes
5	Send malformed JSON	HTTP 400; body {"status":400,"code":"VALIDATION_ERROR","message":"The request body is not valid JSON or i	Yes
6	Compare row counts (users, sessions, help requests) before and after	before 22,6,6; after 22,6,7 (one new help request expected)	Yes

TC-16: Audit log (NFR-06, SR-07): Pass

Expected: 3 records with user, time and action; edit and delete are refused

#	Step	Observed	OK
1	Read the audit log for the logins, booking and Admin action done above	4 matching records; kinds found: ['BOOKING', 'LOGIN', 'SUSPEND']; example: LOGIN	Yes
2	Check each record has user, time and action	columns: id, user_id, action, timestamp	Yes
3	Edit a record as the application's database user	ERROR: permission denied for table audit_log	Yes
4	Delete a record as the application's	ERROR: permission denied for table	Yes

#	Step	Observed	OK
	database user	audit_log	
5	Edit a record as the database owner	ERROR: audit_log is append-only	Yes

TC-17: Dashboard and responsive layout (FR-13, NFR-05): Pass

Expected: Upcoming sessions and pending requests (and a Tutor's rating) are shown; no horizontal scroll; at most 3 clicks

#	Step	Observed	OK
1	Open the Learner dashboard at 360 px wide	horizontal overflow: 0 px; bottom tab bar visible: True; tabs: Home, Find, Sessions, Groups, Alerts	Yes
2	Dashboard shows upcoming sessions and pending requests (FR-13)	'Upcoming sessions' section: True; 'Waiting for a reply' section: True	Yes
3	Find a tutor page at 360 px	horizontal overflow: 0 px	Yes
4	Book a slot from the Learner dashboard	3 clicks (Find a tutor, Show matching tutors, a time); confirmation: "Request sent to Meera Iyer. They have 48 hours to reply."; booked with Meera Iyer	Yes
5	Tutor (Meera Iyer) accepts from the dashboard at 360 px	1 click; average rating shown on dashboard: True; horizontal overflow: 0 px	Yes
6	Open the Learner dashboard at 1280 px wide	horizontal overflow: 0 px; side navigation visible: True	Yes

Note: Run in Chromium (Playwright) at 360 x 740 and 1280 x 800. Screenshots tc17-1 to tc17-7.

TC-18: API response time (NFR-02): Pass

Expected: At least 95% of responses take under 1 second

#	Step	Observed	OK
1	100 virtual users (each logged in) call GET /api/dashboard and POST /api/requests for 60 seconds through nginx over HTTPS (backend/scripts/load-test.js)	requests=9053 ok=9053 p50=8ms p95=29ms 100.0% of responses were successful and under 1 second (target: 95%) -> PASS	Yes

Note: Pacing keeps each user near 90 requests a minute, under the 100-per-minute limit of NFR-04.

TC-19: HTTPS and token expiry (SR-01, SR-04): Pass

Expected: Plain HTTP is rejected or redirected and TLS is 1.2 or higher; the old token gets 401 TOKEN_EXPIRED; no token gets 401

#	Step	Observed	OK
1	Request http://localhost:8080/api/users/me (plain HTTP)	HTTP 301, Location: https://localhost/api/users/me	Yes
2	Connect with TLS 1.2 (openssl s_client -tls1_2)	New, TLSv1.2, Cipher is ECDHE-RSA-AES256-GCM-SHA384; Protocol : TLSv1.2	Yes
3	Connect with TLS 1.1 (client forced to allow it)	40C74875057F0000:error:0A00042E:SSL routines:ssl3_read_bytes:tlsv1 alert protocol version:../ssl/record/rec_layer_s3.c:1599:SSL alert number 70; New, (NONE), Cipher is (NONE)	Yes
4	Call GET /api/users/me with a token issued 3 hours ago	HTTP 401 TOKEN_EXPIRED	Yes
5	Call GET /api/users/me with no token	HTTP 401 UNAUTHENTICATED	Yes
6	Call GET /api/users/me with a malformed token	HTTP 401 UNAUTHENTICATED	Yes

TC-20: Deployed stack on Kubernetes (SR-01, FR-02): Pass

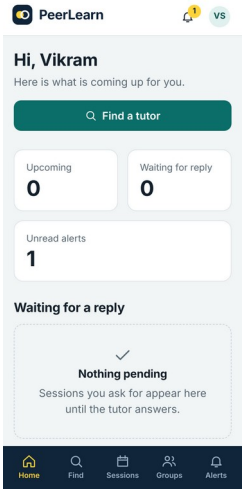
Expected: All pods ready and the migration Job complete; the home page returns PeerLearn; login returns a token

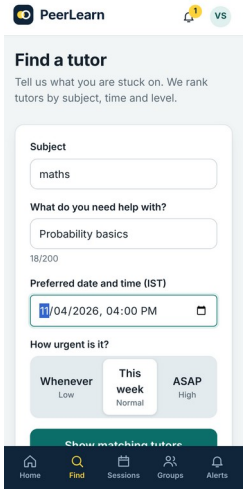
#	Step	Observed	OK
1	Push to main (merge of PR #33, commit 32855fc) starts the CI workflow; jobs backend and frontend run the tests and build	backend: success; frontend: success	Yes
2	Job docker builds both images and pushes them to Docker Hub (amoghshetty05/peerlearn-api and amoghshetty05/peerlearn-web, tagged with the commit)	docker: success (04:38:16 to 04:38:57 UTC, 31 Oct 2026)	Yes
3	Job kubernetes creates a kind cluster and runs k8s/deploy.sh with the images just pushed	deployment db, api and web successfully rolled out; job.batch/migrate condition met; pods api 1/1, db 1/1, web 2 of 2 Running	Yes
4	Open the home page and log in as Admin through the web Service over HTTPS	home page returned PeerLearn; login returned a token: “Frontend and API respond through Kubernetes”	Yes

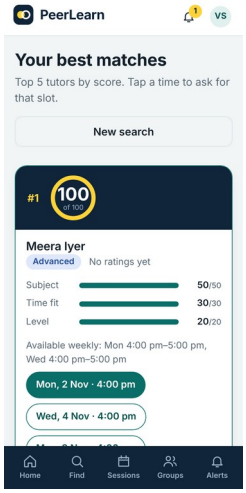
Note: Run automatically in GitHub Actions: https://github.com/amoghshetty05/peerlearn/actions/runs/41169331310 (job kubernetes).

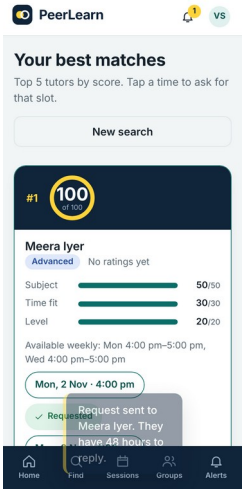
4. Screenshots (TC-17)

Phone width, 360 px: the Learner dashboard, the Find a tutor form, the ranked matches, and the booking confirmation.

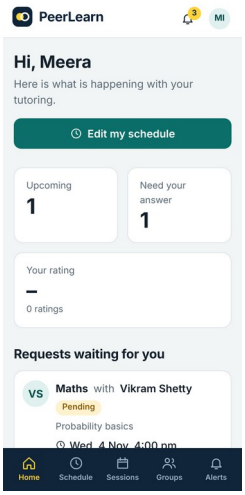


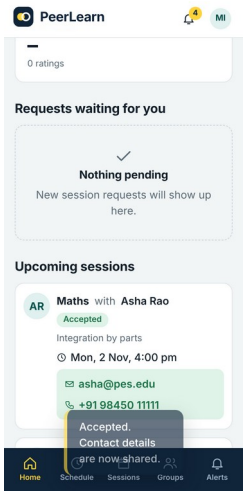




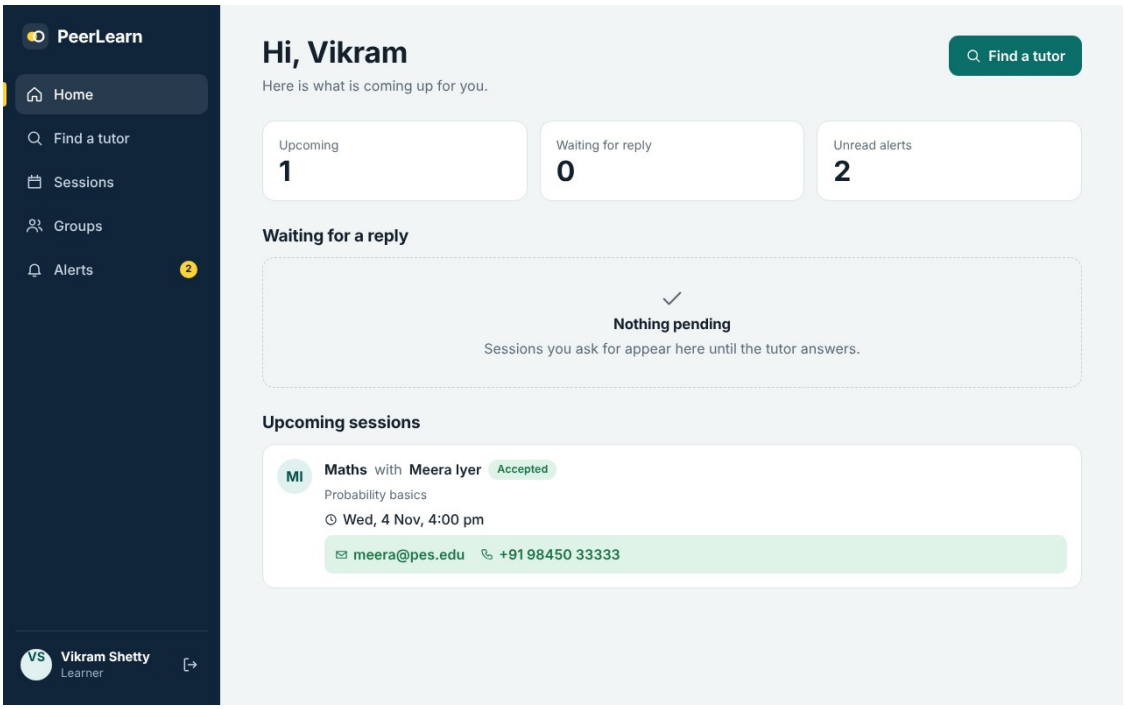


The Tutor's dashboard before and after accepting with one click (360 px):





The Learner dashboard at desktop width, 1280 px, after the Tutor accepted: contact details are now shown (FR-14).



5. Requirement coverage

Requirement	System test cases	Result
FR-01	TC-01, TC-02	Pass
FR-02	TC-03, TC-04, TC-20	Pass
FR-03	TC-05	Pass
FR-04	TC-05	Pass
FR-05	TC-06	Pass
FR-06	TC-06	Pass
FR-07	TC-06	Pass
FR-08	TC-07, TC-08	Pass
FR-09	TC-09	Pass
FR-10	TC-10	Pass
FR-11	TC-07, TC-08	Pass
FR-12	TC-11	Pass
FR-13	TC-09, TC-17	Pass
FR-14	TC-07	Pass
FR-15	TC-06	Pass
NFR-01	TC-12	Pass
NFR-02	TC-18	Pass
NFR-03	TC-13	Pass

Requirement	System test cases	Result
NFR-04	TC-14	Pass
NFR-05	TC-17	Pass
NFR-06	TC-16	Pass
NFR-07	Not testable here	The system is not hosted, so there is no monthly uptime record (Test Plan section 4)
NFR-08	Unit Test Report	Pass: 94.9% line coverage against a 70% target
SR-01	TC-19, TC-20	Pass
SR-02	TC-12	Pass
SR-03	TC-07	Pass
SR-04	TC-03, TC-19	Pass
SR-05	TC-04	Pass
SR-06	TC-15	Pass
SR-07	TC-16	Pass
SR-08	TC-14	Pass

6. Defects and observations

Defects found: none.

Observations (not defects):

1. The HTTP-to-HTTPS redirect keeps the host name but not a non-standard port, so on the test ports (8080 and 8443) it pointed to port 443. In the real deployment both ports are standard (80 and 443), where the redirect is correct.
2. TC-08 needs 48 hours to pass. The request's timestamps were moved 49 hours back in the test database and the unchanged background job then expired it within 55 seconds.
3. TC-06 needed a Tutor with three ratings. These were inserted directly into the test database as test data; TC-09 separately shows a rating made through the system itself.